

CPT-Symmetric Kähler–Dirac Fermions

./logos/cambridge-cropped.pdf application in Cosmology & Fundamental Physics

Latham Boyle & Wei-Ning Deng — Higgs Centre, Edinburgh & Cavendish/KICC, Cambridge



CPT-Symmetric Univ.

figures/CPT_periodic.png

Big Bang = CPT mirror
⇒ solves cosmological
const, flatness problem,
dark matter (ν_R), scale
invariant PPS, **No inflation**

**But WHY must the Big Bang
be a CPT mirror?**

Kähler–Dirac Fermions

lattice.png

KD fermions are 4D p -forms:
$$\Phi = \sum \varphi_{\mu_1 \dots \mu_p} dx^{\mu_1} \wedge \dots \wedge dx^{\mu_p}$$
$$(K - m)\Phi = (i\gamma^\mu \partial_\mu - m)\Psi = 0$$

Live naturally on the lattice ⇒
solve fermion doubling prob.

Ghost Problem & Sol.

KD_Majorana.png

Spinor form Ψ (4×4 matrix):
half the columns have **wrong-**
sign Lagrangian ⇒ negative-
norm states (ghosts)!

Sol: wrong sign = PT reversal

The KD-Majorana Condition: Field Theory Forces CPT

Black Mirror [arXiv:2412.09558]

KD-Majorana condition

$\psi^c \equiv (i\gamma^2)\psi^*(i\gamma^2) = \psi$
Forces every particle on sheet 1
to pair with its anti-particle on
sheet 2 — **CPT is built in.**

SM Unification

1 SM generation = 4 KD fields
 $\mathcal{L}_{\text{KD}} = \text{Tr}[\bar{\Psi}_A(i\not{D})\Psi_A],$
 $\mathcal{L}_Y = \text{Tr}[\mathcal{H}^T \hat{\Psi}'^i_{A,L} \Psi'^j_{A,R} \mathcal{Y}^{ij}_A]$
 ν_R **required by geometry.**

arXiv:2511.11548

annihilate_BHmirror.png

BH horizon = PT mirror.
Infalling particles **annihilate**
with anti-partners at the horizon.
(KD-Maj. guarantees every particle

Sounds like...

tenet_2.jpg

Same:
T-reversed anti-universe,
BHs as transition gates.

Different:
you & anti-you
annihilate at the gate!